



PIMPRI CHINCHWAD EDUCATION TRUST'S
PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
(An Autonomous Institute)
Affiliated to Savitribai Phule Pune University (SPPU)
ISO 21001:2018 Certified by TUV SUD

T.Y. B. TECH

Year: 2025 – 26

Subject: FSDL

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Branch: Computer Engineering

Division: B

Assignment:6

1. Objective

The primary objective of this assignment is to design and develop an online e-commerce portal for buying and selling used vehicles such as cars and bikes using Node.js and Express.js. The goal is to create a dynamic and user-friendly platform where users can browse listings, view detailed product information, and book appointments or express interest in purchasing vehicles.

The application utilizes a NoSQL database like MongoDB to efficiently store and manage vehicle data, user details, and booking information. The project also aims to demonstrate how backend technologies can be integrated with a modern frontend interface to build a complete, scalable web application.

2. Learning Outcomes

After completing this assignment, the following skills and knowledge are gained:

- Understanding backend development using Node.js and Express.js.
 - Learning how to design and implement RESTful APIs.
 - Gaining experience in working with NoSQL databases like MongoDB.
 - Understanding how to manage product listings dynamically.
 - Learning how to implement features like search, filtering, and booking/appointment systems.
 - Improving UI/UX design for e-commerce platforms.
 - Enhancing full-stack development and database integration skills.
-

3. Brief Theory

An e-commerce portal for used items (such as cars or bikes) is a web-based application that allows users to buy, sell, or explore second-hand products. These platforms rely on dynamic data handling and efficient database management to provide real-time information.

Node.js is used as the server-side runtime environment, allowing JavaScript to be executed on the backend. Express.js simplifies server creation, routing, and middleware handling, making it easier to build scalable web applications.

MongoDB, a NoSQL database, stores data in a flexible JSON-like format. This is ideal for e-commerce applications where product details (such as price, model, images, and specifications) may vary. It allows fast data retrieval and easy scalability.

In this project (CarBazaar), the system works as follows:

- Vehicle data (cars/bikes) including images, price, model, and features are stored in the MongoDB database.
- The backend server fetches and sends this data to the frontend dynamically.
- Users can browse featured deals, search for specific vehicles, and view detailed listings.
- The application allows users to book appointments or show interest in a vehicle.
- The system processes these requests and stores booking/user interaction data for further use.

This workflow demonstrates how real-world e-commerce platforms manage listings, user interactions, and backend processing efficiently.

Code:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Bookings — CarBazaar</title>

<link rel="stylesheet" href="/css/style.css">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;800&family=Racing+Sans+One&display=swap" rel="stylesheet">

</head>
```

```

<body>

<nav class="navbar">

  <a href="/" class="logo">  <span>Car</span>Bazaar</a>

  <div class="nav-links">

    <a href="/">Browse</a>

    <a href="/new.html">Sell Your Car</a>

    <a href="/bookings.html">Bookings</a>

    <a href="/cart.html" class="cart-link">  Cart <span class="badge" id="cartCount">0</span></a>

  </div>

</nav>


<section class="section">

  <h1 class="section-title">  All Bookings</h1>

  <div id="bookingsList" class="bookings"></div>

  <div id="emptyBookings" class="empty" style="display:none">No bookings yet.</div>

</section>


<footer><p>© 2026 CarBazaar</p></footer>


<script src="/js/app.js"></script>

<script>

(async () => {

  const bookings = await fetch('/api/bookings').then(r => r.json());

  const list = document.getElementById('bookingsList');

  if (!bookings.length) { document.getElementById('emptyBookings').style.display = 'block'; return; }

  list.innerHTML = bookings.map(b => `

    <div class="booking-card">

      <div class="booking-img" style="background-image:url('${b.car ? b.car.image : ''}')"></div>

      <div>

        <h3>${b.car ? b.car.brand + ' ' + b.car.model : 'Deleted car'}</h3>

        <p><b>${b.buyerName}</b> · ${b.buyerEmail} · ${b.buyerPhone}</p>

```

```

<p class="meta">${b.message || ""}</p>

<span class="status status-${b.status.toLowerCase()}">${b.status}</span>

<span class="meta"> · ${new Date(b.createdAt).toLocaleString()}</span>

</div>

</div>`).join("");

})();

</script>

</body>

</html>

```

```

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Car Details — CarBazaar</title>

<link rel="stylesheet" href="/css/style.css">

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;800&family=Racing+Sans+One&display=swap" rel="stylesheet">

</head>

<body>

<nav class="navbar">

<a href="/" class="logo">  <span>Car</span>Bazaar</a>

<div class="nav-links">

<a href="/">Browse</a>

<a href="/new.html">Sell Your Car</a>

<a href="/bookings.html">Bookings</a>

<a href="/cart.html" class="cart-link">  Cart <span class="badge" id="cartCount">0</span></a>

</div>

</nav>

```

```
<section class="detail" id="detail"></section>
```

```
<section id="book" class="section">
```

```
<h2 class="section-title">&img alt="calendar icon" data-bbox="348 154 371 169"/> Book a Test Drive / Inquiry</h2>
```

```
<form id="bookingForm" class="form-card">
```

```
<input name="buyerName" placeholder="Your Name" required>
```

```
<input name="buyerEmail" type="email" placeholder="Email" required>
```

```
<input name="buyerPhone" placeholder="Phone" required>
```

```
<textarea name="message" placeholder="Any special requests..." rows="3"></textarea>
```

```
<button class="btn btn-primary" type="submit">Confirm Booking</button>
```

```
</form>
```

```
</section>
```

```
<section class="section">
```

```
<h2 class="section-title">Similar Cars</h2>
```

```
<div class="grid" id="similarGrid"></div>
```

```
</section>
```

```
<footer><p>© 2026 CarBazaar</p></footer>
```

```
<script src="/js/app.js"></script>
```

```
<script src="/js/detail.js"></script>
```

```
</body>
```

```
</html>
```

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Cart — CarBazaar</title>
```

```
<link rel="stylesheet" href="/css/style.css">
```

```

<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;800&family=Racing+Sans+One&display=swap" rel="stylesheet">

</head>

<body>

<nav class="navbar">

  <a href="/" class="logo">  <span>Car</span>Bazaar</a>

  <div class="nav-links">

    <a href="/">Browse</a>

    <a href="/new.html">Sell Your Car</a>

    <a href="/bookings.html">Bookings</a>

    <a href="/cart.html" class="cart-link">  Cart <span class="badge" id="cartCount">0</span></a>

  </div>

</nav>

<section class="section">

  <h1 class="section-title">  Your Garage Cart</h1>

  <div id="cartList" class="cart-list"></div>

  <div id="cartTotal" class="cart-total" style="display:none">

    <h2>Total: ₹<span id="totalVal">0</span></h2>

    <button class="btn btn-ghost" id="clearBtn">Clear Cart</button>

    <button class="btn btn-primary" onclick="alert('  Checkout flow coming soon!')">Checkout</button>

  </div>

  <div id="emptyCart" class="empty" style="display:none">Your cart is empty. <a href="/">Browse cars →</a></div>

</section>

<footer><p>© 2026 CarBazaar</p></footer>

<script src="/js/app.js"></script>

<script>

```

```

async function loadCart() {
  const r = await fetch('/api/cart');
  const { cars, total } = await r.json();
  const list = document.getElementById('cartList');
  if (!cars.length) {
    document.getElementById('emptyCart').style.display = 'block';
    document.getElementById('cartTotal').style.display = 'none';
    list.innerHTML = "";
    return;
  }
  document.getElementById('emptyCart').style.display = 'none';
  document.getElementById('cartTotal').style.display = 'block';
  document.getElementById('totalVal').textContent = total.toLocaleString('en-IN');
  list.innerHTML = cars.map(c => `
    <div class="cart-item">
      <div class="cart-img" style="background-image:url('${c.image}')"></div>
      <div class="cart-info">
        <h3>${c.brand} ${c.model}</h3>
        <p class="meta">${c.year} · ${c.fuel} · ${c.location || ""}</p>
        <p class="price">₹${c.price.toLocaleString('en-IN')}</p>
      </div>
      <button class="btn btn-danger" onclick="removeItem('${c._id}')">Remove</button>
    </div>`
  ).join("");
}

async function removeItem(id) {
  await fetch('/api/cart/' + id, { method: 'DELETE' });
  loadCart(); updateCartCount();
}

document.getElementById('clearBtn').addEventListener('click', async () => {
  await fetch('/api/cart', { method: 'DELETE' });
  loadCart(); updateCartCount();
});

```

```
});  
loadCart();  
</script>  
</body>  
</html>  
  
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<title>CarBazaar — Premium Used Cars</title>  
<link rel="stylesheet" href="/css/style.css">  
<link  
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;800&family=Racing+Sans+One&display=swap" rel="stylesheet">  
</head>  
<body>  
<nav class="navbar">  
  <a href="/" class="logo">  <span>Car</span>Bazaar</a>  
  <div class="nav-links">  
    <a href="/">Browse</a>  
    <a href="/new.html">Sell Your Car</a>  
    <a href="/bookings.html">Bookings</a>  
    <a href="/cart.html" class="cart-link">  Cart <span class="badge" id="cartCount">0</span></a>  
  </div>  
</nav>  
  
<section class="hero">  
  <div class="hero-content">  
    <h1 class="hero-title">Find Your <span class="gradient">Dream Ride</span></h1>
```



```
<p class="hero-sub">India's premium marketplace for certified pre-owned cars 🚀 </p>

<form id="searchForm" class="search-bar">

  <input type="text" id="searchInput" placeholder="Search by brand or model...">

  <button type="submit">🔍 Search</button>

</form>

</div>

<div class="hero-blob"></div>

</section>

<section class="section">

  <h2 class="section-title">★ Featured Deals</h2>

  <div class="grid" id="featuredGrid"></div>

</section>

<section class="section">

  <div class="section-head">

    <h2 class="section-title">🚗 All Cars (<span id="carCount">0</span>)</h2>

    <div class="filters">

      <select id="filterBrand"><option value="">All Brands</option></select>

      <select id="filterFuel">

        <option value="">All Fuels</option>

        <option>Petrol</option><option>Diesel</option><option>Electric</option><option>Hybrid</opt
ion><option>CNG</option>

      </select>

      <select id="filterSort">

        <option value="">Sort: Latest</option>

        <option value="low">Price: Low → High</option>

        <option value="high">Price: High → Low</option>

        <option value="new">Year: Newest</option>

      </select>

    </div>

  </div>
```

</div>

<div class="grid" id="carGrid"></div>

<div class="empty" id="emptyState" style="display:none">☹ No cars found. Be the first to list one!</div>

</section>

<footer><p>© 2026 CarBazaar — Drive your dream. Built with Node.js + Express + MongoDB.</p></footer>

<script src="/js/app.js"></script>

<script src="/js/home.js"></script>

</body>

</html>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Sell Your Car — CarBazaar</title>

<link rel="stylesheet" href="/css/style.css">

<link

href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;800&family=Racing+Sans+One&display=swap" rel="stylesheet">

</head>

<body>

<nav class="navbar">

  CarBazaar

<div class="nav-links">

Browse

Sell Your Car

Bookings

```

    <a href="/cart.html" class="cart-link"><img alt="Shopping Cart icon" data-bbox="455 68 475 85"/> Cart <span class="badge" id="cartCount">0</span></a>
  </div>
</nav>

<section class="section">
  <h1 class="section-title"><img alt="Car icon" data-bbox="350 215 370 230"/> Sell Your Car</h1>
  <form id="newForm" class="form-card big-form">
    <div class="row">
      <input name="brand" placeholder="Brand (e.g. Toyota)" required>
      <input name="model" placeholder="Model (e.g. Fortuner)" required>
    </div>
    <div class="row">
      <input name="year" type="number" placeholder="Year" required>
      <input name="price" type="number" placeholder="Price (₹)" required>
    </div>
    <div class="row">
      <input name="kmDriven" type="number" placeholder="KM Driven">
      <input name="color" placeholder="Color">
    </div>
    <div class="row">
      <select
name="fuel"><option>Petrol</option><option>Diesel</option><option>Electric</option><option>Hy
brid</option><option>CNG</option></select>
      <select name="transmission"><option>Manual</option><option>Automatic</option></select>
    </div>
    <input name="location" placeholder="Location">
    <input name="image" placeholder="Image URL">
    <textarea name="description" placeholder="Describe your car..." rows="4"></textarea>
    <button class="btn btn-primary" type="submit">List Car</button>
  </form>
</section>

```

```
<footer><p>© 2026 CarBazaar</p></footer>
```

```
<script src="/js/app.js"></script>
```

```
<script>
```

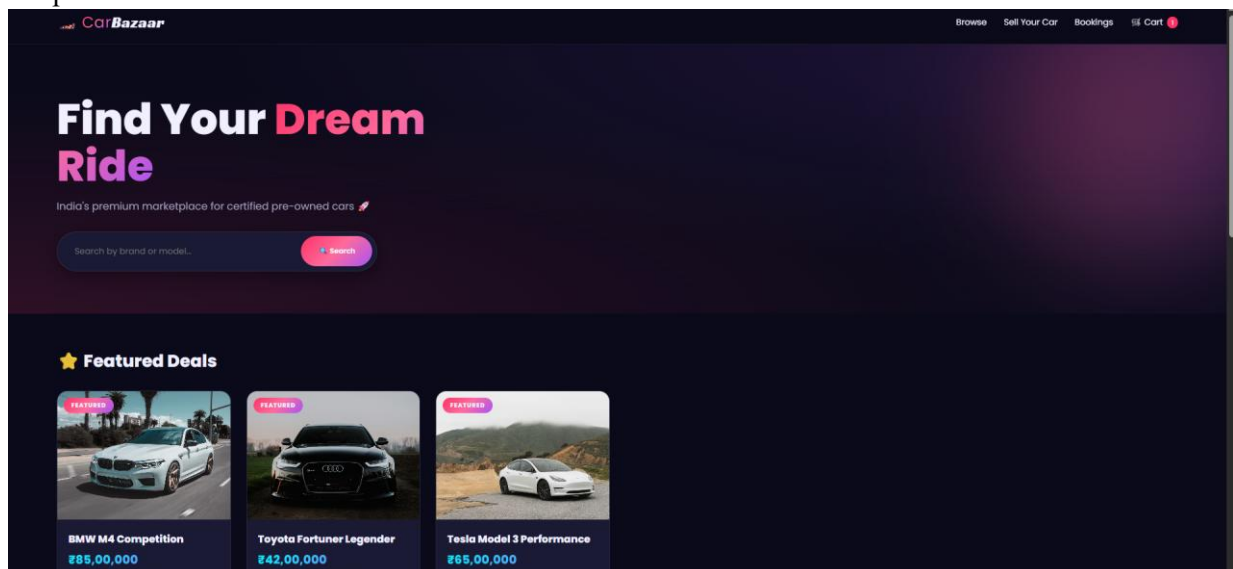
```
document.getElementById('newForm').addEventListener('submit', async e => {  
  e.preventDefault();  
  const data = Object.fromEntries(new FormData(e.target));  
  const res = await fetch('/api/cars', {  
    method: 'POST', headers: {'Content-Type': 'application/json'}, body: JSON.stringify(data)  
  });  
  const car = await res.json();  
  location.href = '/car.html?id=' + car._id;  
});
```

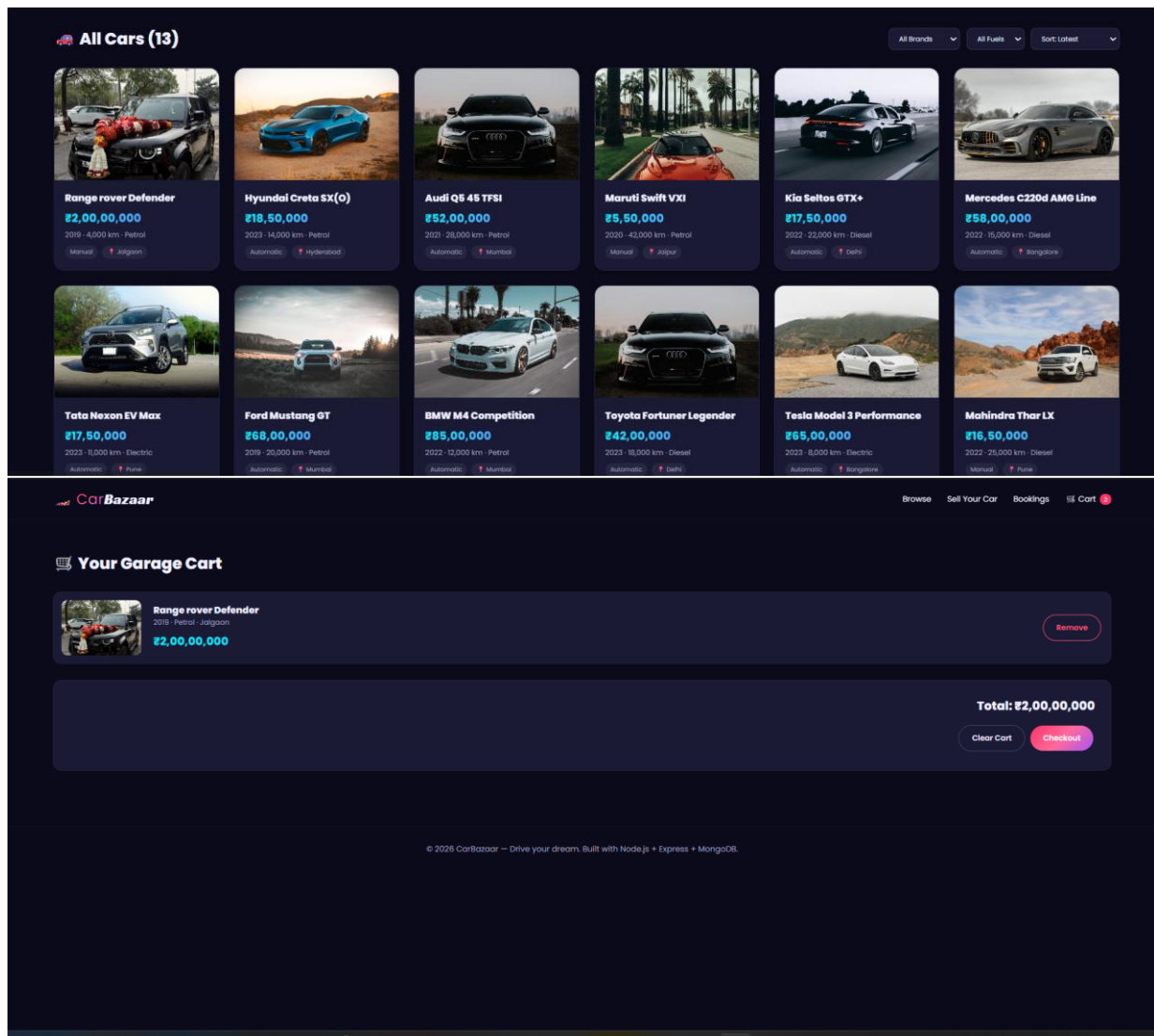
```
</script>
```

```
</body>
```

```
</html>
```

Output:





Conclusion:

In this assignment, a functional e-commerce portal for used vehicle sales was successfully developed using Node.js, Express.js, and MongoDB. The project demonstrates the implementation of dynamic product listings, search functionality, and appointment booking features.

The integration of backend logic with a responsive frontend interface ensures a smooth and interactive user experience. This project provides practical exposure to real-world application development and highlights the importance of database management and server-side processing.

Overall, the assignment strengthens full-stack development skills and provides a solid foundation for building scalable and user-centric web applications.